Experiment-19

19. CBC Encryption Using 3DES

Aim:

To write a C program to encrypt plaintext using Triple DES in CBC mode.

Procedure:

1. Read a 24-byte 3DES key.
2. Initialize the CBC initialization vector (IV).
3. Read the plaintext and add padding.
4. Encrypt the data using 3DES in CBC mode.
5. Display the ciphertext in hexadecimal.

Code (q19.c)

```
#include <stdio.h>
#include <string.h>
#include <openssl/des.h>

int main(void) {
    DES_cblock k1, k2, k3, iv = {0};
    DES_key_schedule s1, s2, s3;
    unsigned char data[1024];
    char key[49];
    int i, n, pad;

    printf("Enter 48 hex digits for 3DES key: ");
    scanf("%48s", key);

    for (i = 0; i < 8; i++) {
        unsigned int x;
        sscanf(key + 2*i, "%2x", &x); k1[i] = x;
        sscanf(key + 16 + 2*i, "%2x", &x); k2[i] = x;
        sscanf(key + 32 + 2*i, "%2x", &x); k3[i] = x;
    }

    DES_set_key_unchecked(&k1, &s1);
    DES_set_key_unchecked(&k2, &s2);
```

```
    DES_set_key_unchecked(&k3, &s3);

    printf("Enter plaintext: ");
    getchar();
    if (!fgets((char *)data, sizeof(data), stdin)) return 1;

    n = (int)strlen((char *)data);
    if (n > 0 && data[n-1] == '\n') n--;

    pad = 8 - n % 8;
    for (i = 0; i < pad; i++) data[n+i] = (unsigned char)pad;
    n += pad;

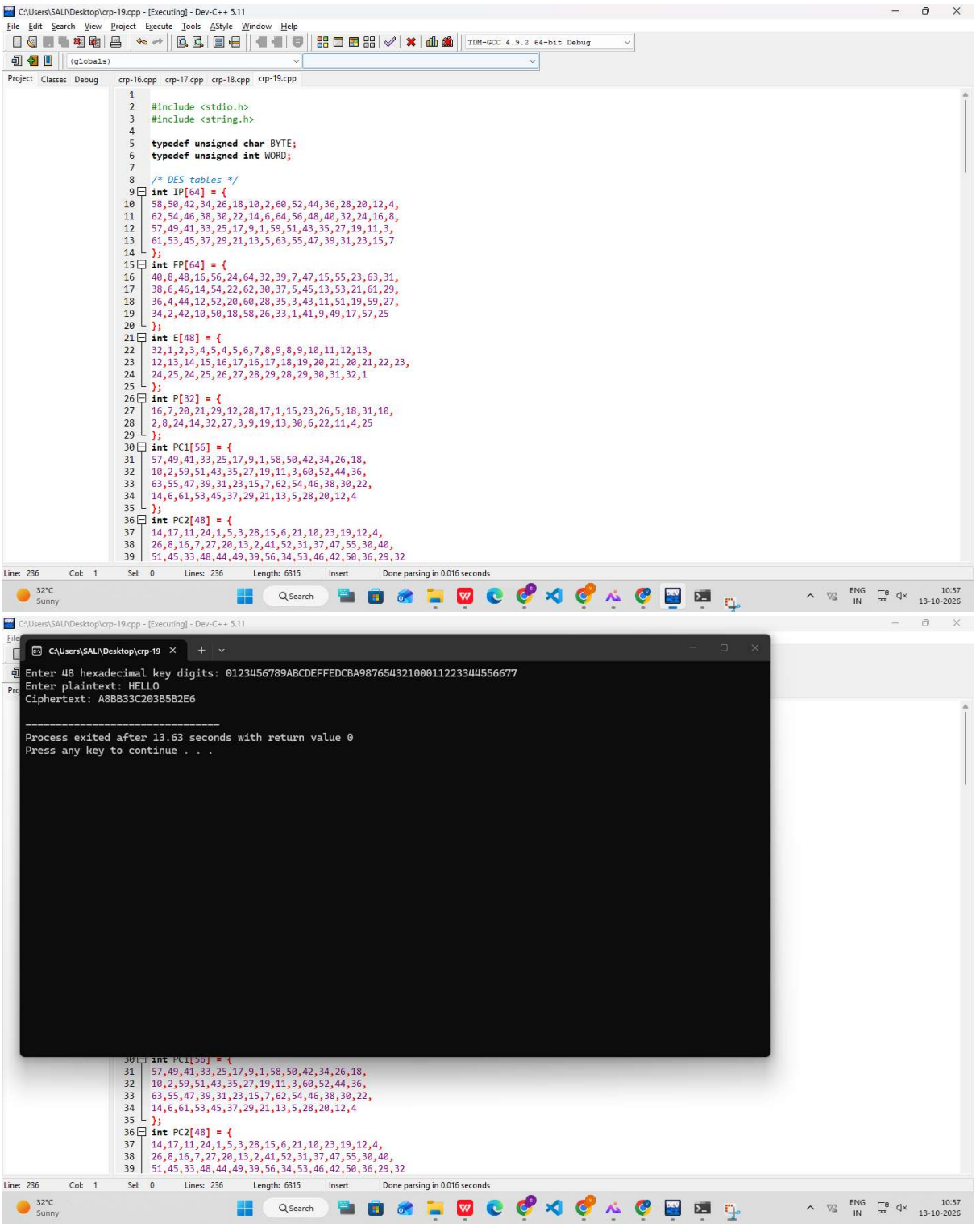
    DES_ede3_cbc_encrypt(data, data, n, &s1, &s2, &s3,
                         &iv, DES_ENCRYPT);

    printf("Ciphertext: ");
    for (i = 0; i < n; i++) printf("%02X", data[i]);
    printf("\n");
    return 0;
}
```

Sample Input

```
Enter 48 hex digits for 3DES key:
0123456789ABCDEFFEDCBA987654321089ABCDEF01234567
Enter plaintext: HELLO
```

Sample Output



Result:

The plaintext is encrypted using 3DES in CBC mode. AES is recommended over 3DES for modern applications.